

Phase-dependent modulation of human corticospinal plasticity by associative pairing of transcranial and neuromuscular stimulation

Phase-dependent rhythmic neuromodulation of plasticity

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Abstract

Communication in the brain is mediated by oscillatory synchronization. Both the oscillatory power and phase determine effective signal transmission. They moderate the responsiveness and specificity of neural activity to incoming information. Their impact on activity-dependent plasticity in the human brain is, however, unknown.

Male and female participants volitionally modulated the sensorimotor oscillatory power by imagined finger extension. The corresponding real movement was artificially induced by peripheral stimulation of the forearm muscle. Importantly, the timing of the peripheral input with respect to cortical oscillations was systematically modified. For this purpose, a novel phase-dependent rhythmic stimulation protocol was established. Transcranial alternating current stimulation (TACS) of the motor cortex and neuromuscular electrical stimulation (NMES) of the forearm were applied simultaneously at the same frequency (i.e., 20 Hz). In randomized order, the peripheral stimuli were given in one of four equidistant TACS-driven cortical phases, and without TACS. Corticospinal excitability (CSE) changes following the intervention were indexed with transcranial magnetic stimulation. NMES induced a 20 Hz rhythmic entrainment over the sensorimotor cortex. Associative NMES/TACS led to lasting bidirectional plasticity, i.e., increases (+157.5%) and decreases (-60.82%) of CSE compared to the baseline depending on the timing of the pairing. CSE was modulated in a sinusoidal manner and increased only when the peripheral input arrived at the rising flank of the cortical beta phase.

In the same responsive brain state, the direction and extent of corticospinal plasticity is determined by phase-specific input. This may be relevant when establishing closed-loop interventions for state-dependent therapeutic purposes, e.g., after stroke.

Significance Statement

Power and phase of oscillatory synchronization moderate the responsiveness and specificity of the human brain to incoming information. Their impact on activity-dependent plasticity is, however, unknown. Participants volitionally modulated the power of sensorimotor oscillations by imagined finger extension. The corresponding real movement was induced by peripheral stimulation of the forearm muscle, while the sensorimotor oscillations were synchronized by cortical stimulation, both at 20 Hz. When the timing of the peripheral input was systematically modified with respect to the cortical phase, bidirectional plasticity was revealed. Corticospinal excitability increased only when the peripheral input arrived at the rising flank of the cortical oscillation. In the same responsive brain state, the direction and extent of corticospinal plasticity is determined by phase-specific input.

Introduction

Oscillatory synchronization of neurons has been proposed to mediate communication in the brain. Both the power and phase of these oscillations determine effective signal transmission. Theoretical concepts, such as gating by inhibition (Jensen and Mazaheri, 2010), and communication through coherence (Fries, 2015), may explain the power- and phase-dependent neuronal interactions that open windows of increased responsiveness. From an integrative perspective (Bonnefond et al., 2017), oscillatory power and phase moderate the sensitivity and specificity of neuronal populations to incoming information.

In the human motor system, empirical support for this conceptual framework has been provided by applying single-pulse transcranial magnetic stimulation (TMS) over the primary motor cortex. Oscillatory power fluctuations and phase-specific gain modulation in the beta frequency band influenced corticospinal excitability (CSE) independently of each other, as shown by amplitude changes of motor evoked potentials (MEP) in the corresponding muscle in both the resting and active motor system (Khademi et al., 2018; Naros et al., 2019). When investigating the instantaneous influence of the oscillatory power on CSE, low as opposed to high beta power was observed to lead to a 40–70% excitability increase over a wide range of power values. By contrast, the phase modulation was critically dependent on the precise timing of the stimuli to the rising phase of the oscillatory beta cycle, but could then achieve CSE increases of 180% (Khademi et al., 2019).

In our aim to develop a state-dependent intervention to augment long-lasting CSE changes, brain-technology interfaces constitute a powerful tool for modulating brain states and synchronizing them with cortical or peripheral input (Kraus et al., 2016, Biasiucci et al., 2018). Specifically, brain state-dependent TMS controlled by volitional modulation of the sensorimotor beta-band induced a robust CSE increase (Kraus et al., 2016). By contrast,

when applied independent of the brain state, the same stimulation pattern resulted in a decrease of corticospinal excitability. When sensorimotor alpha oscillations were used to trigger TMS, targeting the oscillatory trough vs. peak led to an increase vs. to no change in CSE in a preselected group with high sensorimotor alpha power (Zrenner et al., 2018). However, when the same approach was applied in non-selected individuals, this finding was not replicated, suggesting that alpha activity may be primarily implicated in attentional gating rather than in causing a generalized modulation of cortical excitability (Madsen et al., 2019). By contrast, when transcranial alternating current stimulation (TACS) rhythmically modulated sensorimotor beta oscillations at 20 Hz, short-interval intracortical inhibition and intracortical facilitation changed in a phase-dependent manner (Guerra et al., 2016). Corticospinal propagation became accordingly less variable and more rapid at the optimum phase of beta oscillations (Torrecillos et al., 2020).

In a complementary line of research in non-human primates, short-term bidirectional synaptic plasticity of cortico-cortical connections caused by invasive cortical stimulation outlived the conditioning paradigm for 1-2 seconds, when applied phase-locked during sensorimotor beta oscillations (Zanos et al., 2018). Furthermore, when combining phase-locked stimulation with neural operant conditioning, local networks were shifted between oscillatory and non-oscillatory states (Peles et al., 2020). It remains unclear, however, as to whether phase-dependent stimulation along the beta oscillatory cycle would also lead to longer-lasting changes of more distant, polysynaptic connections, and whether this could be achieved non-invasively in the human brain.

We therefore developed a novel intervention protocol on the basis of associative cortical beta entrainment to investigate the hypothesis that phase-specific input determines the direction and extent of corticospinal plasticity. Specifically, participants volitionally

modulated the power of sensorimotor oscillations by imagined finger extension. The corresponding real movement was induced by peripheral stimulation of the forearm muscle, while the sensorimotor oscillations were synchronized by cortical stimulation, both via rhythmic stimulation in the beta frequency band (i.e., 20 Hz). We systematically varied the timing of the peripheral input with respect to the cortical phase, and found evidence for bidirectional plasticity, with a maximal increase of corticospinal excitability when the peripheral input arrived at the rising flank of the cortical oscillation.

Materials and Methods

All subjects gave their written informed consent before participation in the study which was approved by the ethics committee of the medical faculty of the University of Tübingen. A total of 24 subjects participated in 66 experimental sessions.

Phase-dependent NMES/TACS

In 14 right-handed subjects (eight female, age $M = 25.5$, $SD = 3.48$), cortical beta oscillations were driven by 20 Hz TACS to the right motor cortex during MI of finger and wrist extension while phase-dependent 20 Hz NMES was concurrently applied to the corresponding left forearm muscles (i.e., extensor digitorum communis, EDC). The subjects participated in four interventions which were performed during four different sessions, each separated by at least two days. The interventions differed in the specific timing of the associative NMES/TACS pairing. The 20 Hz NMES was triggered at one specific phase of the 20 Hz sine signal of the transcranial stimulator during each of the four interventions. The order of the four sessions, i.e., phases, was randomized for each subject. The targeted phases were equidistant and identical across subjects. We triggered stimulation at either 35° , 125° , 215° or 305° of the TACS signal. These phases were chosen since they exhibit unique voltage values and are equidistant, thus facilitating a proper reconstruction of a sinusoidal modulation. During each session, additional tasks and measurements, as specified below, were performed before and after the intervention (see *Experimental Design*).

Time course and topography of cortical evoked responses following NMES

In ten right-handed subjects (six female, age $M = 24.6$, $SD = 2.8$), the time course and topography of the evoked cortical responses were assessed in one session by applying the same NMES intervention as used in the phase-dependent study, but without

concurrent TACS. The subjects received the same tasks and measurements prior to and following the intervention. Since TACS electrodes were not placed or stimulated, we could measure the cortical response induced by NMES in the absence of TACS effects or artifacts of electrodes and stimulation (Noury et al., 2016). It also provided us with the opportunity to estimate plastic effects in the absence of paired stimulation.

Experimental design

Each session adhered to the same procedure (Figure 1A). During the intervention period of approximately 40 min, a total of 165 stimulation periods, consisting of 3 s bursts of concurrent NMES/TACS, were applied. This resulted in a cumulative stimulation period of 8.25 min consisting of 10065 NMES pulses per intervention (see *Intervention*). Prior to and following the intervention, TMS and NMES measurements were performed to investigate the cortical and spinal excitability changes, respectively. Participants performed a task for depotentiation of fragile stimulation effects (Todd et al., 2009; Goldsworthy et al., 2015) and to reduce plastic after-effects of TMS and NMES measurements (Touge et al., 2001). This depotentiation task was based on contracting the EDC muscle and maintaining this extension for ten seconds while targeting 30% relative to maximum voluntary contraction (MVC). Subjects received visual feedback of their contraction level on a computer screen. Each contraction trial was repeated ten times per task. The depotentiation task was performed after the first TMS measurement, after each NMES measurement, and directly after the intervention. The TMS measurement used to estimate the plastic changes was preceded by two depotentiation tasks and performed 30 min after the intervention. We therefore consider the ensuing TMS measurements as indicative of robust and long-term plastic changes.

Electrophysiological evaluation

We applied 64 active electrodes with the actiCAP EEG system (Brain Products GmbH) and a BrainAmp DC amplifier with a sampling rate of 1000 Hz. The following electrode channels were recorded and analyzed: Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, TP9, CP5, CP1, CP2, CP6, TP10, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10, AF7, AF3, AF4, AF8, F5, F1, F2, F6, FT9, FT7, FC3, FC4, FT8, FT10, C5, C1, C2, C6, TP7, CP3, CPz, CP4, TP8, P5, P1, P2, P6, PO7, PO3, POz, PO4, and PO8. AFz and FCz were used as the ground and reference electrodes, respectively. Impedance was reduced with SuperVisc electrolyte gel (Easycap, Herrsching). Electromyography (EMG) was recorded with Neuroline 720 (AmbuGmbH) surface electrodes which were placed over the left and right EDC muscles in a belly-belly montage and amplified with BrainAmp ExG.

Transcranial magnetic stimulation

Transcranial Magnetic Stimulation (TMS) was applied over the primary motor cortex with an MCF-B70 Butterfly Coil (MagPro X100/R30 magnetic stimulator; MagVenture GmbH) in combination with the TMS Navigator (Localite GmbH) neuronavigation software. Biphasic stimulation was provided in posterior-anterior coil orientation and at a 45° angle to the longitudinal fissure. The procedure consisted of approximately 100 pulses for hotspot detection and 70 pulses for acquiring a stimulus-response curve. The hotspot for the left EDC was detected as follows: First, approximately 40 stimuli with an intensity of 40% of the maximum stimulator output (MSO) were applied in the vicinity of the anatomical hand knob. If these 40 stimuli did not suffice to elicit any MEPs at the initial intensity, the latter was increased by 5% of MSO, and the procedure was repeated. Once MEPs were elicited, we detected the three locations with the highest peak-to-peak amplitudes using the 3D visualization in the neuronavigation software. Each of these positions was stimulated a further three times. The position with the highest average MEP value was defined as the

hotspot. In all measurements, any EDC muscle preactivation was ruled out by visual inspection and automatic prevention of stimulation whenever the rectified background EMG activity exceeded 20 μ V. The resting motor threshold (RMT) was determined as the stimulation intensity resulting in 5 out of 10 MEPs above 50 μ V. The stimulus-response curve was assessed from 90% to 150% of the RMT in steps of 10%. Each of the intensities was applied in a block of 10 stimuli. Blocks were applied in a randomly permuted order. The MEP amplitude was measured as the peak-to-peak amplitude (V_{pp}) from 15 to 50 ms after the stimulus.

Neuromuscular electrical stimulation

Neuromuscular Electrical Stimulation (NMES) was applied over the left EDC (STG 4008; Multi Channel Systems GmbH) with surface electrodes (Neuroline 720; Ambu GmbH). The procedure for the acquisition of a stimulus-response curve consisted of 80 pulses. Ten biphasic pulses (a positive rectangular pulse with a width of 500 μ s followed by a negative pulse of identical width) were applied with a uniformly jittered inter-stimulus interval (ISI) of 1 to 2 s and at eight different intensities (2, 4, 6, 8, 10, 12, 14, or 16mA). All stimuli were administered in a randomly permuted order. We measured the peak-to-peak amplitude (V_{pp}) of the EMG response of the left EDC in the time period from 5 to 35 ms after each stimulus. We then fitted a three-parameter sigmoidal model to the EMG response amplitudes (V_{pp}) over stimulation intensities I , with parameter A representing the maximal response at saturation intensity, σ the slope at the inflection point, and θ the location of the inflection point on the intensity line.

$$1. \quad V_{pp} = \frac{A}{1 + \exp(-\sigma + (I - \theta))}$$

Transcranial alternating current stimulation

Transcranial alternating current stimulation (TACS) was applied to the primary motor cortex at a stimulation intensity of 1 mA peak-to-peak (neuroConn DC PLUS stimulator, neuroCare, Ilmenau) by placing a round carbon-rubber electrode on C4 (thickness 1 mm, outer diameter 7.5 cm, inner diameter 2 cm). A rectangular return electrode was placed in posterior-anterior orientation on Pz (thickness 3 mm, shape 5 cm × 7 cm). The impedance was reduced to below 10 kΩ (Ten20 Conductive Neurodiagnostic Electrode Paste; Weaver and Company). Where necessary, additional electrolyte gel (SuperVisc) was applied for impedance reduction. The electrical field for constant current stimulation with this montage was simulated with ROAST (Huang et al., 2019).

Intervention

The intervention consisted of 15 runs, each of which contained 11 trials, resulting in a total of 165 trials. Each trial was comprised of a 2 s preparation period, a 6 s motor imagery (MI) period and a 6 s rest period (see figure 1B). Subjects were instructed to perform kinesthetic motor imagery of left-hand opening (i.e., extension of fingers and wrist) during the motor imagery period, and to relax during the rest period. The onset of each period was signified by an auditory cue. Stimulation was initiated 1 to 1.5 s after the onset of the MI period, with a uniform random jitter to reduce habituation. We applied 60 cycles of TACS at 20 Hz, with one fade-in and one fade-out cycle, resulting in 3 s of stimulation at full amplitude. The post-stimulation period was adapted in accordance with the pre-stimulation period to maintain a 6 s motor imagery period (see figure 1B). The TACS device controlled the concurrent NMES by sending a trigger-out signal at each cycle, which was routed to the peripheral stimulator and the amplifier via a microcontroller. In each trial, 61 peripheral pulses were applied (one after the fade-in, 60 after each cycle). This resulted in a total of 10065 phase-dependent peripheral pulses for each intervention,

and 9900 full TACS cycles (see figure 1D). NMES was applied as a biphasic pulse, beginning with a positive peak of 500 μ s width and followed by a negative peak of 500 μ s width (see figure 1C). Each subject received NMES at a stimulation intensity for the individual inflection point θ rounded to the nearest 0.1 mA. This intensity was strong enough to induce a pronounced extension of the fingers and wrist (see figure 1E). The TACS stimulation showed that the electrical field was spanned predominantly between the two electrodes covering the primary and secondary sensorimotor areas in the right hemisphere (see figure 1F).

Analysis of evoked responses

The cortical response evoked by NMES pulses was assessed as follows: We selected all EEG channels and isolated the signal for the period from 1s before to 1s after each stimulus, resulting in 80 trials per subject. All trials were centered and re-referenced on the basis of a common average. Artifacts EEG channels were processed by spherical spline interpolation with Fieldtrip (Oostenveld et al., 2011). To further increase robustness against noise, we took the median of all trials in each stimulation intensity. This approach resulted in a data tensor of 64 channels x 2000 time points x 8 intensities for each subject. We calculated the global mean field potential (GMFP) as the standard deviation across channels. To ascertain whether the GMFP response was affected by the stimulation intensity, we fitted a sigmoidal model (as described in 1.) to each time point and compared it against a constant model. We then selected the period during which the GMFP exhibited a significant sigmoidal modulation to analyze the cortical topography of the response (see figure 2A). The latency from the peripheral stimulation to the cortex was determined as follows: Each subject's GMFP was averaged across all intensities and smoothed with a Gaussian kernel of 5 ms width. In each subject, we determined two latencies for every response by taking the time point at which the respective GMFP peaked. We used the

latencies of all 10 subjects to estimate the parameters of the log normal distributions for each component. This enabled us to probabilistically model the timing of the transient cortical activity induced by peripheral stimulation (see figures 2 & 3).

Analysis of phase-dependent effects

The phase-dependency of CSE changes was assessed by fitting a sinus on the pre-post TMS differences in accordance with the following formula, with C being the constant main effect of the intervention, A the amplitude of the sinus, and ϕ its phase-delay.

$$2. \delta_{prepost} = C + (A \times \sin(x - \phi))$$

The amplitude A is a measure for the strength of the sinusoidal modulation. However, since it can only be positive, even for pure noise, it shows a bias. To protect our analysis from this bias, we therefore assessed the statistical significance of the fit using a permutation test with 1000 repetitions. By testing the statistical significance of the fit, we could ascertain whether the modulation of plasticity was phase dependent. Furthermore, a sinus peaks once per cycle. The fitting therefore enabled us to analytically determine the phase of the TACS at which the NMES should be triggered so as to induce the maximal plastic increase.

Data availability

Data and code are available upon request.

Results

Neuromuscular electrical stimulation effects

During the intervention, NMES was performed at the intensity of the inflection point of the stimulus-response curve, i.e., on average with 9.03 mA (with a 95% confidence interval ranging from 8.25 to 9.81 mA). The location of the inflection point for the NMES stimulus-response curve did not differ between the groups with and without TACS ($t(22) = -1.27$, $p = 0.22$), or between the four phase-dependent sessions ($F(3, 52) = 0.35$, $p = 0.79$). There were no amplitude differences of the muscle response to NMES in the pre- and post-intervention measurements (all $p > 0.01$ with threshold Bonferroni-corrected for multiple comparison at 0.007, permutation test with 1000 repetitions), indicating that the corticospinal plasticity observed was not related to changes of spinal excitability.

Cortical entrainment with neuromuscular electrical stimulation

Single NMES pulses (during the pre- and post-intervention measurements) evoked two cortical responses. The first response peaked on average at around 80 ms, and was significantly elevated from 49 to 100 ms after the stimulus (see figure 2A). It exhibited a parietal-to-frontal dipole across the sensorimotor cortex in the right hemisphere (see figure 2B). The second response peaked on average at around 205 ms, and showed significant elevations between 108 to 246 ms after the stimulus (see figure 2A). Its central topography indicated a radial dipole orientation (see figure 2C). The model for rhythmic entrainment (with repetitive NMES during the intervention) suggested that only the first cortical response induced a sinusoidal oscillation of cortical activity at 20 Hz (see figure 2D), while the second exhibited only limited modulation and accumulated towards a plateau (see figure 2E).

Corticospinal plasticity with phase-dependent rhythmic stimulation

In the NMES condition without TACS, no changes in corticospinal excitability were observed across any of the seven TMS intensities ($p > 0.19$). When NMES was combined with TACS, the corticospinal excitability was modulated in a phase-dependent manner at 100% RMT. Bidirectional plasticity, with robust and long-lasting increases and decreases of corticospinal excitability ensued, depending on the timing of the NMES/TACS pairing (see figure 3A). The observed sinusoidal modulation of corticospinal plasticity was significant after correction for multiple comparisons (permutation test with 1000 repetitions, $p = 0.004$) with deflections at the respective phases of 215° ($M = 157.50\%$, $p = 0.004$), 305° ($M = -41.19\%$, $p = 0.4$), 35° ($M = -60.82\%$, $p = 0.084$), and 125° ($M = 5.41\%$, $p = 0.644$).

After fitting the sinusoidal model, the estimated optimal TACS phase for triggering NMES to achieve a maximal increase of corticospinal plasticity was $\sim 203^\circ$ ($CI_{95\%} = 127^\circ - 239^\circ$; see figure 3B). The offset of the sine was $\sim 15\%$ with a peak-to-peak amplitude of $\sim 223\%$, and a $\sim 126\%$ excitability increase compared to baseline.

Triggering NMES at the estimated optimal TACS phase of $\sim 203^\circ$ would probabilistically induce peaks in cortical activity at the TACS phase of $\sim 24^\circ$ for the first (fronto-parietal) and at $\sim 40^\circ$ for the second (centro-radial) component, when taking the latencies and entrainment effects into account (see figure 3C). Accordingly, the optimal phase (for external input arriving at the cortex) to bring about plastic changes corresponded to the rising flank of 20 Hz oscillations (see figure 3D).

To estimate the inter-subject reliability of the findings, we calculated the individual optimal trigger phase for each subject. In 10 of the 14 subjects, the highest increase of corticospinal plasticity was found to occur when the afferent input reached the rising flank of cortical beta oscillations. This effect was significant in a binomial test with $p = 0.03$.

Discussion

We combined neuromuscular electrical stimulation (NMES) and transcranial alternating current stimulation (TACS) for rhythmic alignment of proprioceptive input to the oscillatory state of the human sensorimotor cortex. This approach provided new insight on the phase-dependent integration of long-latency, polysynaptic afferent input into the sensorimotor system, and the plastic influence on motor output. The key finding of this study is the detection of an optimal phase to induce a robust and long-lasting plastic increase of human corticospinal excitability. This effect could be attributed to the arrival of the afferent input during the rising flank of cortical beta oscillations. Notably, corticospinal excitability was modulated in a sinusoidal manner and revealed bidirectional plasticity.

NMES pulses induced two broad and long-latency cortical responses, such as were previously described for potentials evoked by proprioception (Arnfred et al., 2000; Seiss et al., 2002). These responses could be distinguished by latency and topography and exhibited different steady state dynamics during 20Hz-NMES (see figure 2). The first cortical response maintained a rhythmic perturbation at a periodicity of 20Hz, while the second response accumulated towards a plateau. These findings complemented recent observations during 20Hz transcutaneous stimulation of the upper arm at rest. This intervention led to cortical entrainment at the stimulation frequency also, which was attributed to short-latency somatosensory afferences (Asamoah et al., 2019).

The cortical entrainment induced in our study was driven by long-latency, polysynaptic afferences; NMES induced functional finger and wrist extension, and stimulated the brain during motor imagery, i.e., an active motor system (Stinear et al., 2005), which was relevant for two reasons: First, polysynaptic afferent input bears a closer resemblance to the condition of natural movement than pure sensory stimulation. Second, in accordance

with the concept of active inference, motor output is considered to be sensitive to the proprioceptive context (Friston et al., 2009). Manipulation of this context may, therefore, affect motor cortex efferences (Gandolla et al., 2014). Specifically, by applying a rhythmic entrainment paradigm, we could control with high precision and on a short time-scale the arrival of the proprioceptive afferent input relative to the state of the active motor cortex.

To ensure that the proprioceptive input to the cortex arrived with the necessary temporal precision during specific oscillatory states, these states were entrained by TACS. The effects of TACS in the human motor system are well documented (Pogosyan et al., 2009; Joundi et al., 2012), even though the underlying mechanisms, particularly at low stimulation intensities, are still under research (Liu et al., 2018). Probable mechanisms of action for TACS are the rhythmic modulation of excitability due to depolarization and hyperpolarization, direct frequency-specific enhancement of the natural resonance characteristics of the underlying local neural circuits (Zaehle et al., 2010; Guerra et al., 2016) and/or the indirect modulation of cortical neurons via rhythmic activity from peripheral nerves (Asamoah et al., 2019). In any case, the effects of rhythmic cortical stimulation appear to be state dependent. Specifically, rhythmic entrainment was stronger during weak endogenous oscillations, since these appeared to be more malleable by the low stimulation intensity of TACS (Alagapan et al., 2016). In this study, the responsiveness of the motor cortex to external perturbation was maximized in three ways: We stimulated the motor cortex at 20Hz to modulate intrinsic resonant oscillations (Pogosyan et al., 2009; Romei et al., 2016), reduced the influence of endogenous sensorimotor oscillatory power in this frequency band via a motor imagery task (Bauer et al., 2015; Vukelic and Gharabaghi, 2015), and applied concurrent NMES (Reynolds et al., 2015).

The present study detected changes in CSE which lasted for at least 30min after the intervention and which were robust against repeated depotentiation. This suggested that

the observed modulation of corticospinal plasticity was similar to long-term potentiation /depression-like plasticity (Bear and Malenka, 1994). Furthermore, our study showed that the intervention effects were phase dependent. Earlier research on oscillatory power-dependent stimulation (Klein-Flügge et al., 2013; Royter and Gharabaghi, 2016; Kraus et al., 2018) observed robust increases of corticospinal excitability of about 20%. In the present study, we ascertained that pairing afferent input along the phase of the cortical beta rhythm led to CSE increases of more than 150%.

Notably, the sinusoidal plastic change was linked to the arrival of afferent input at specific phases of the transcranial stimulation. Earlier work showed that the immediate MEP amplitude in response to TMS was sinusoidally modulated by ongoing endogenous beta oscillations (Khademi et al., 2018; Torrecillos et al., 2020). As in our study, this effect was observed at a TMS intensity of 100% RMT, and the maximal increase was reached when the input arrived at the rising flank of the beta oscillations. One interpretation of these observations is, therefore, that plastic changes (as shown in the present study) and acute modulation of corticospinal excitability (as detected earlier) are linked to the same neuronal population in the motor cortex, which is sensitive to phase-dependent input. Repeatedly targeting this population at the same oscillatory phase, as performed in our intervention, seems to generalize their sensitivity to a later weak external input, even if this input is applied independently of a specific phase.

The specificity of the observed plasticity effect for near-motor threshold stimulation (i.e., 100% RMT) may be explained in accordance with the potential mechanisms of phase-dependent gain modulation (Khademi et al., 2018): Probe stimulation must be weak if it is to be influenced by the synchronized phase patterns of alternating, often sub-threshold, up and down states of neuronal populations; if stimulation is supra-threshold, it will inevitably trigger responses regardless of whether neurons are in relative up or down states. Due to

maximum response variability, the phase-dependent modulation was, therefore, revealed only near threshold (Klein-Flügge et al., 2013).

One complementary explanation is related to the activation of distinct neuronal circuits with different stimulation intensities (Di Lazzaro et al., 1998). 20Hz-TACS has been shown to influence the excitatory and inhibitory neural circuits in the motor cortex in different ways (Guerra et al., 2016): There is evidence of a phase-independent loss of cholinergic short-latency afferent inhibition (SAI), phase-dependent modulation of glutaminergic intracortical facilitation (ICF), and a frequency- and phase-dependent modulation of gamma-aminobutyric acid (GABA-A)-ergic short-interval intracortical inhibition (SICI). The latter is mainly mediated by late I-waves (Di Lazzaro et al., 1998) and modulates plastic effects of paired associate stimulation (PAS) (Elahi et al., 2012; Ni et al., 2019). This suggests that a link may exist between SICI, late I-waves and the plastic effect identified in our study.

Moreover, transcranial stimulation is believed to act via weak subliminal polarization (Reato et al., 2013; Fröhlich, 2014; Lafon et al., 2017). In this regard, it is similar to short intracortical facilitation (SICF), which is linked to early I-waves and known to interfere with SICI (Peurala et al., 2008). Furthermore, the effect of SICF decreases, while SICI increases when the intensity of the test stimulus is increased (Wagle-Shukla et al., 2009). TMS intensities below 110% RMT induce MEPs by recruiting indirect circuits in the motor cortex, i.e., the early presynaptic activation of the corticospinal pathway (Garry and Thomson, 2009; Di Lazzaro and Rothwell, 2014). The fact that we detected our plastic effects at low stimulation intensities (i.e., 100% RMT) suggests that the sinusoidal modulation found in our study was driven by phasic shifts in the balance of early and late presynaptic input of facilitatory and inhibitory interneural populations.

This work was technically restricted by the fact that the induced cortical oscillations could not be directly measured during the application of TACS. While methods to recover parts

of the endogenous signal during stimulation are available (Noury et al., 2016; Kohli and Casson, 2019), these are not applicable when the artifact is phase-locked. Furthermore, we applied TACS only at 20Hz, and could therefore not show any frequency-specificity of the observed effects. However, earlier studies revealed that the gain of cortical motoneurons (Khademi et al., 2018; Naros et al., 2019) and SICI (Guerra et al., 2016) is modulated in a frequency- and phase-dependent manner.

We intervened at four different phases of the sinusoidal rhythm; an approach that is mathematically sufficient to reconstruct a sine-wave. How ICF?

In conclusion, this study has demonstrated that phase-dependent pairing of rhythmic transcranial and neuromuscular electrical stimulation during motor imagery leads to robust and long-lasting corticospinal plasticity in the human motor system. However, there is still some debate as to the relevance of non-sinusoidal waveforms of the beta-rhythm (Cole and Voytek, 2017). Specifically, recent evidence suggests that beta bursts are peaky and asymmetric in shape (Sherman et al., 2016; Little et al., 2018). Assessing more than four phases or non-sinusoidal transcranial stimulation might therefore enable us to estimate and differentiate more complex modulations of corticospinal excitability. Future studies may also address some of the questions that could not be answered by the present work: What effects occur when NMES and/or TACS frequencies do not match? Are stimulation effects mediated in the motor cortex only? Do plasticity effects last for longer periods of time? Is the modulation of cortical excitability paralleled by changes in measures of behavioral performance or interneuronal physiology captured by SICI or ICF?

In conclusion, this study has demonstrated that phase-dependent pairing of rhythmic transcranial and neuromuscular electrical stimulation during motor imagery leads to robust and long-lasting corticospinal plasticity in the human motor system.

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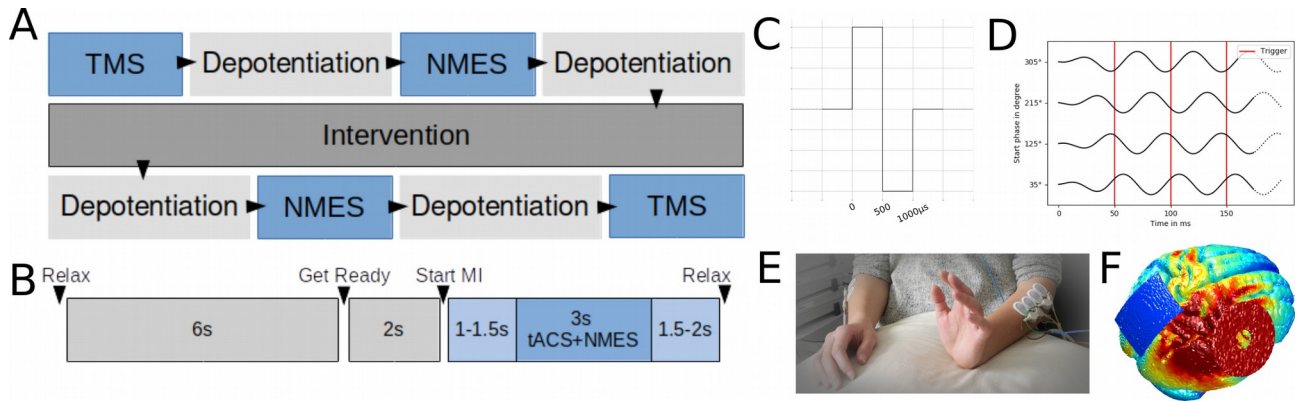


Figure 1: Description of the study design

A Each intervention session consisted of 165 trials of pairing transcranial alternating current stimulation (TACS) and neuromuscular electrical stimulation (NMES), both at 20 Hz. Before and after the intervention, cortical and spinal effects were probed by transcranial magnet stimulation (TMS) and NMES, each of which was followed by a depotentiation task to reduce after-effects of the measurements. Each depotentiation task consisted of ten 10s-contractions of the forearm muscle with 30% of maximum voluntary contraction (MVC). **B** Every trial consisted of a rest, preparation and motor imagery (MI) period. During MI of finger/wrist extension, NMES of the forearm muscle induced the corresponding movement, while TACS was applied to the contralateral motor cortex hand area. **C** NMES pulses had a biphasic rectangular shape. **D** We applied phase-dependent associative NMES/TACS. In each intervention session, NMES was triggered at one of four time-points, i.e., 35°, 125°, 215° or 305° of the TACS cycle. **E** 20 Hz-NMES of the extensor digitorum communis muscle (EDC) for 3s caused finger/wrist extension. **F** The applied TACS montage induced a strong electrical field over sensorimotor areas contralateral to the stimulated forearm.

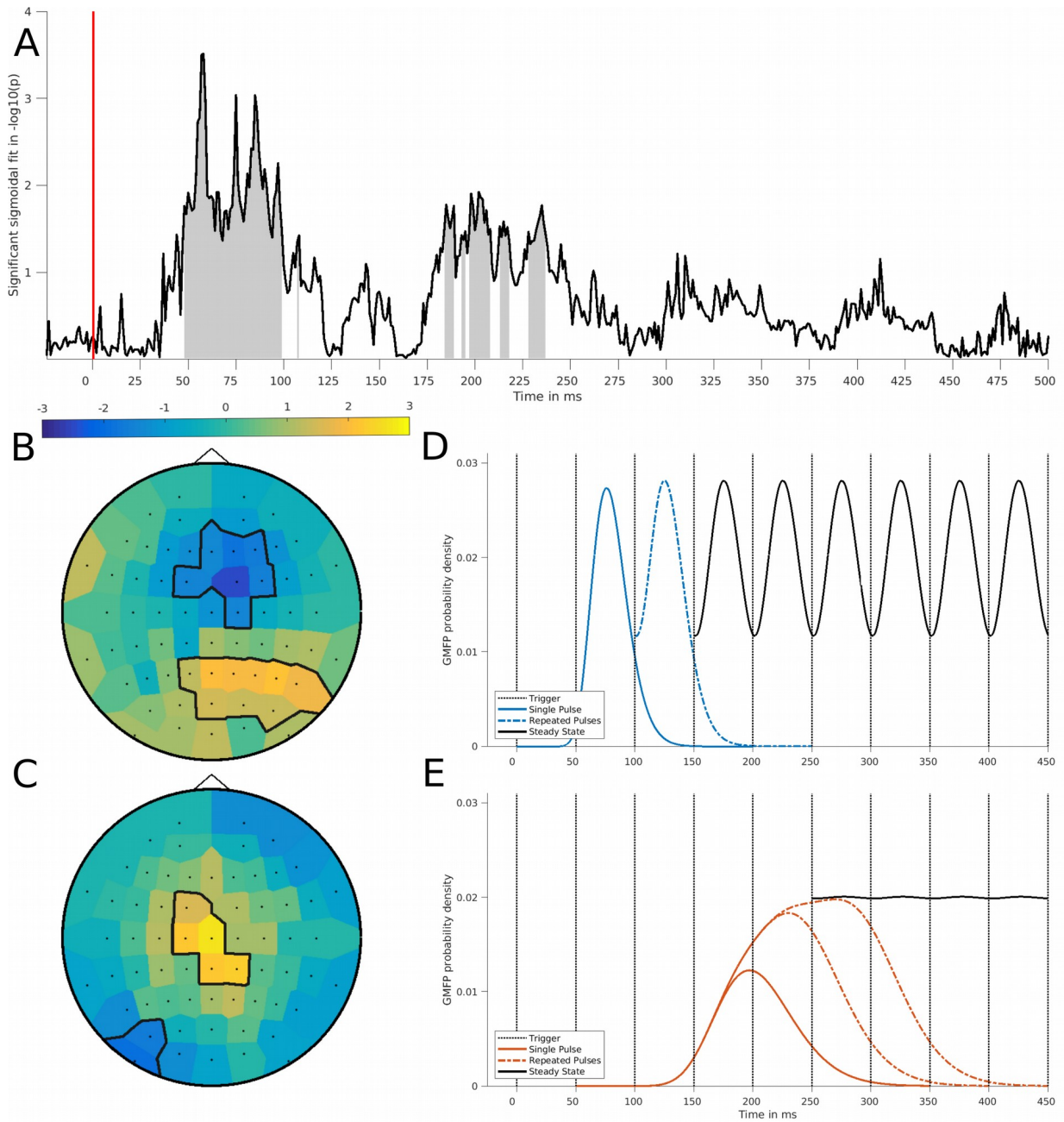


Figure 2: Cortical responses evoked by neuromuscular electrical stimulation (NMES)

A Time course of sigmoidal modulation of the global mean field potential (GMFP) as a function of stimulation intensity of single NMES pulses: x-axis (time); y-axis (statistical significance of the sigmoidal model fit versus a constant model expressed as $[-\log_{10}(p) \times \text{sign of the modulation}]$). Periods where the GMFP was significantly modulated by stimulation intensity are highlighted as gray patches. Single NMES pulses evoked two

cortical responses that peaked at around 80 ms (49-100 ms) and 205 ms (108 - 246 ms) after the stimulus.

B and **C** Exemplary topography of the evoked cortical responses following NMES at an intensity of 16 mA. **B** The first event-related potential (ERP) shows a parietal-to-frontal dipole over the sensorimotor cortex in the right hemisphere, i.e., contralateral to the stimulated forearm. **C** The second ERP shows a central midline topography indicative of a radial dipole orientation. Contours highlight significant deviations of the ERP (color-coded in accordance with the respective color bar) from zero as calculated by a t-test in $[-\log_{10}(p) \times \text{sign of deviation}]$.

D and **E** Time course of the GMFP peak after repetitive NMES, i.e., the temporal probability of a cortical activity burst in response to NMES. The traces show the probability for single pulses (colored trace), a transitional period (interrupted lines), and the steady state during rhythmic NMES entrainment (black traces) of the first and second ERP. The findings suggest that only the former induces a sinusoidal oscillation of cortical activity at 20 Hz (**D**), while the latter exhibits limited modulation and accumulates towards a plateau (**C**).

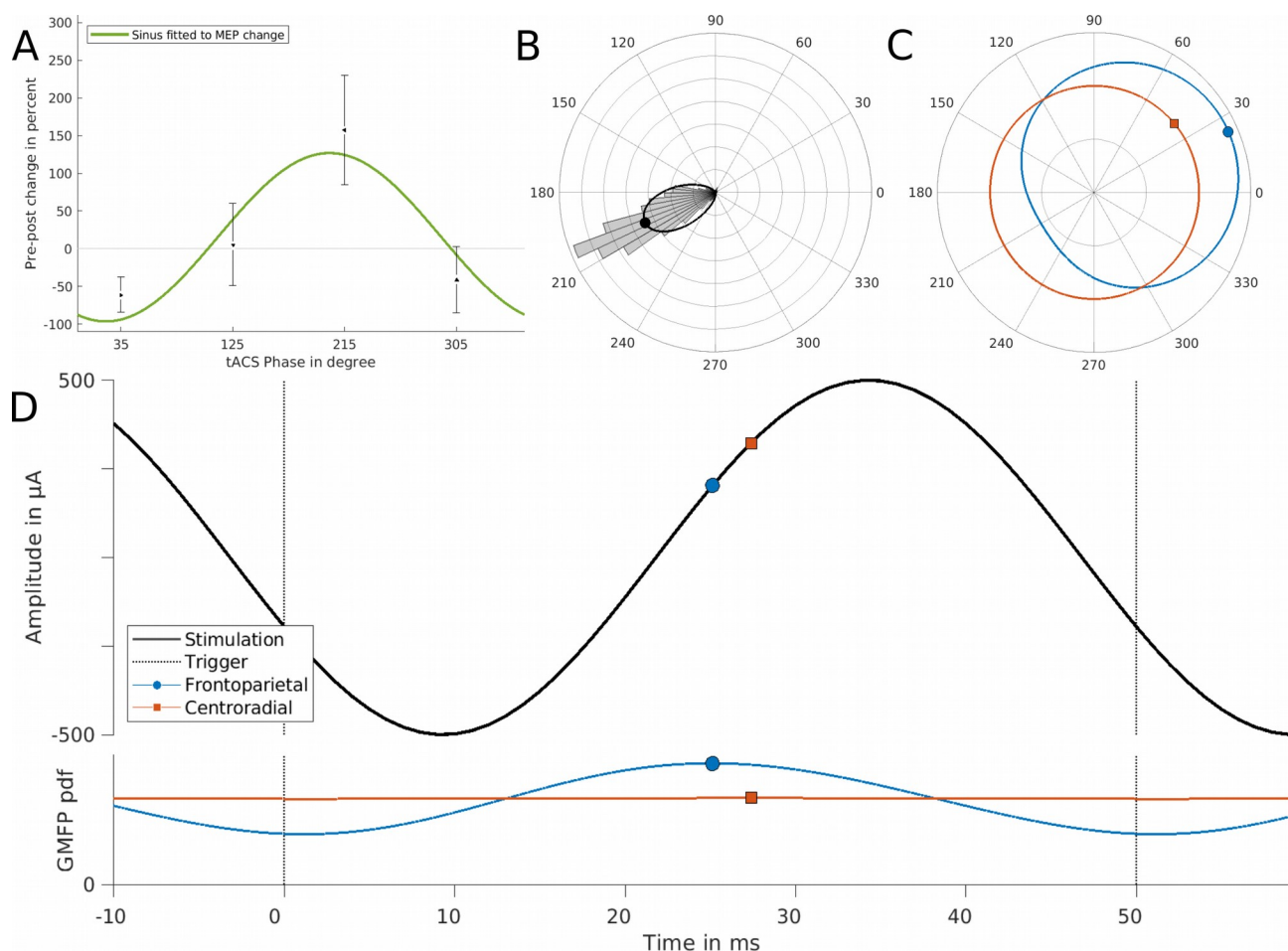


Figure 3: Optimal phase for plastic effects is the rising flank of the 20 Hz oscillation

A Corticospinal plasticity was indexed by robust (following depotentiation) and lasting (+30 min) changes of motor-evoked potentials (MEP) in the target muscle (extensor digitorum communis, EDC) elicited by transcranial magnetic stimulation (TMS) of the contralateral motor cortex. Associative NMES/TACS led to phase-dependent bidirectional plasticity, i.e., increases (+158%) and decreases (-61%) of MEP compared to the baseline motor threshold, depending on the timing of the NMES/TACS pairing. Plasticity was modulated in a sinusoidal manner (green line shows the fitted sinus that was significant after correction for multiple comparisons, permutation test with 1000 repetitions, $p=0.004$). MEP increased only following the intervention condition, where the NMES pulses were triggered at 215° of the TACS phase. Error bars indicate the bootstrapped standard error of the median. The green line shows the sinus fitted to the plastic changes. **B** After fitting the sinusoidal model,

the estimated optimal TACS phase for triggering NMES to achieve a maximal increase of corticospinal plasticity was $\sim 203^\circ$. The bootstrapped distribution of the optimal TACS phase is depicted. The black trace shows the density estimate, and the black dot marks the average optimal TACS phase. **C** Taking the latencies of the rhythmic peaks into account, repetitive NMES applied at the estimated optimal tACS phase would lead to peak cortical activity at $\sim 24^\circ$ for the first (fronto-parietal) and $\sim 40^\circ$ for the second (centro-radial) component. The blue and orange traces indicate the probability density, with the blue circle and orange square marking the phase of maximal probability for the first (fronto-parietal) and second (centro-radial) cortical responses that occur relative to the TACS phase. **D** The linear plot shows the same information as in C. The black trace represents the time course of TACS; the vertical lines indicate NMES pulses triggered at the estimated optimal phase, and the blue and orange traces show the probability density of the first and second cortical responses. The optimal phase for external input to bring about plastic changes corresponded to the rising flank of the 20 Hz oscillation (blue circle).